

INTRODUCTION TO MACROECONOMICS (ECN 1215) LECTURE NOTES



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What is Economics

- Economics is an art or social science that deals with the **allocation of scarce resources to satisfy unlimited human wants**
- Put simply, Economics is a subject that attempts to solve the **scarcity-choice problem**.
- Scarcity – insufficient resources to satisfy all wants fully. i.e. [**Wants are greater than available Resources**]
- Choice – resources allocated to the satisfaction of one want cannot be available to satisfy other wants simultaneously

Cont'

- Economics focuses on the fundamental fact of scarcity – which is the dual existence of insatiable wants and limited resources.
- Economics is the study of how individuals & nations use resources under their command to satisfy their wants fully as possible or to maximize their welfare given their resource constraints.

Economic decision

Three basic questions have to be answered:

1. What goods and services to produce;
 2. How to produce the selected goods and services;
- and
3. For whom – what share of these goods & services are to be given to each individual or household.

Economic decision Cont'

- These questions are also referred to as '*what*', '*how*' and '*for whom*' decisions

Application of Economics in Defence and Security

- The Defence Forces don't have all the resources they need to implement all the programmes they have.
- They are left with no option but MAKE choices

Two Main Branches of Economics

- **Microeconomics**
 - studies behaviour of individual economic units
 - households, firms, markets
- **Macroeconomics**
 - Focus is on the whole/aggregate economy
 - E.g., GNP, GDP, inflation, unemployment

Cont'd

- *Macroeconomics* is about the 'big picture'; it describes and explains the working of the whole economy.
 - *Microeconomics*, as we have seen, is about individual parts that make up the big picture; microeconomics describes and explains the working of the units that make up the whole economy – the household, the business firm and the government.
- Macroeconomics tries to explain the behaviour of the level of prices of all goods and services taken together; the change in this price level from one period to another is known as the *inflation rate*. *Deflation* occurs when the price level falls – a rather infrequent event.
 - The price of one good compared with another good is called the *relative price*; microeconomics attempts to explain relative prices.

Cont'd

- Furthermore, macroeconomics is concerned with the behaviour of total income and total output for the whole economy;
 - microeconomics is concerned with income and output of the individual units that comprise the total economy.
- Macroeconomics studies the employment and unemployment rates for the economy as a whole;
 - microeconomics is concerned with the hiring and firing practices of individual firms.

Economics trains you to. . . .

- ◆ Think in terms of alternatives.
- ◆ Evaluate the cost of individual and social choices.
- ◆ Examine and understand how certain events and issues are related.

The Economist as a Scientist

The economic way of thinking . . .

- ◆ Involves thinking analytically and objectively.
- ◆ Makes use of the scientific method.

The Scientific Method

- ◆ Uses abstract models to help explain how a complex, real world operates.
- ◆ Develops theories, collects, and analyzes data to prove the theories.

Observation, Theory and More Observation!

The Role of Assumptions

- ◆ Economists make assumptions in order to make the world easier to understand.
- ◆ The art in scientific thinking is deciding which assumptions to make.
- ◆ Economists use different assumptions to answer different questions.

The Economic Way of Thinking

- ◆ Includes developing *abstract models* from theories and the analysis of the models.
- ◆ Uses two approaches:
 - ◆ ***Descriptive*** (reporting facts, etc.)
 - ◆ ***Analytical*** (abstract reasoning)

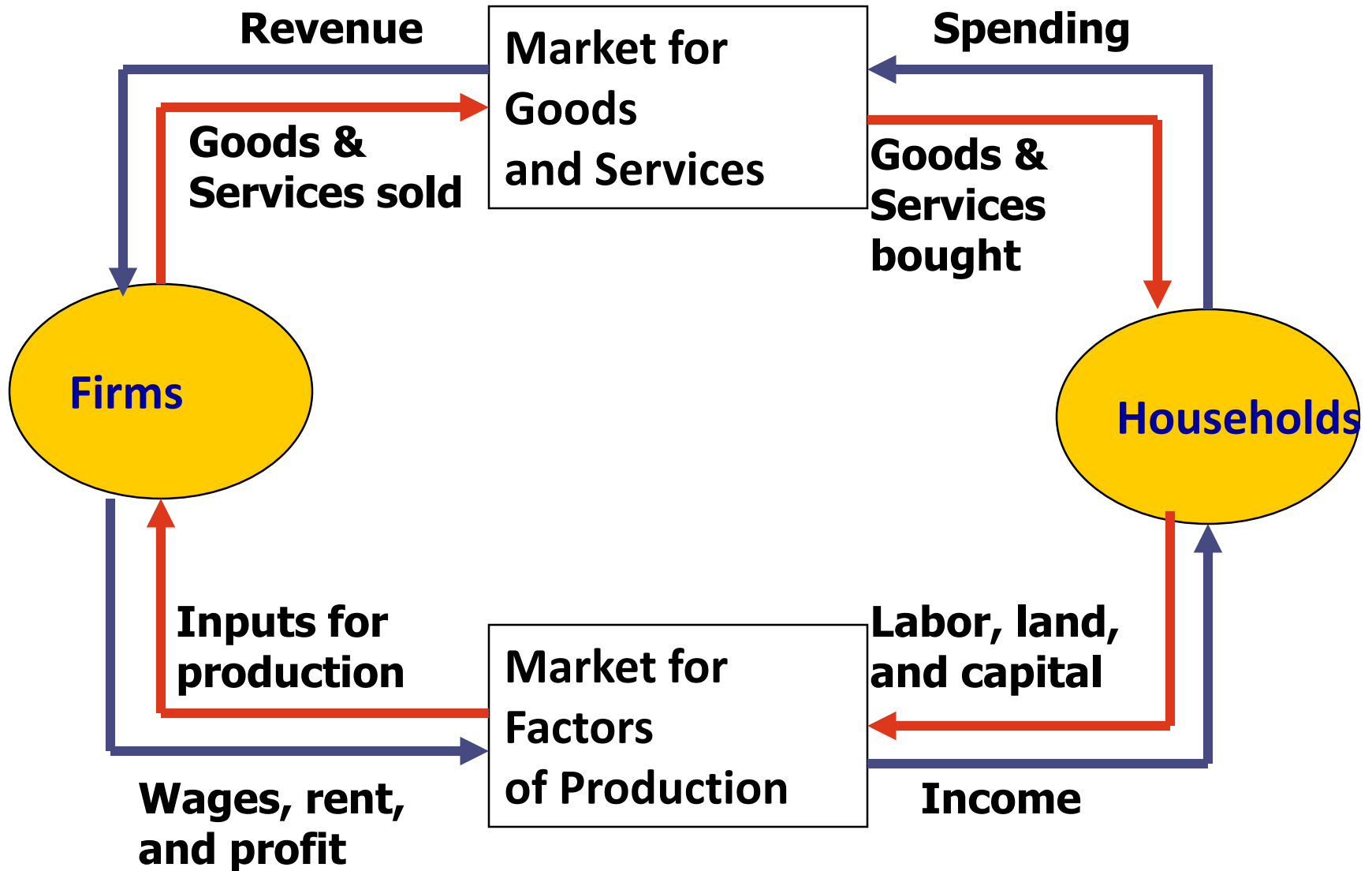
Economic Models

- ◆ Economists use models to simplify reality in order to improve our understanding of the world
- ◆ Two of the most basic economic models include:
 - ◆ *The Circular Flow Model*
 - ◆ *The Production Possibilities Frontier*

The Circular-Flow Model

The circular-flow model is a simple way to visually show the economic transactions that occur between households and firms in the economy.

The Circular-Flow Diagram



The Circular-Flow Diagram

Firms

- ◆ Produce and sell goods and services
- ◆ Hire and use factors of production

Households

- ◆ Buy and consume goods and services
- ◆ Own and sell factors of production

The Circular-Flow Diagram

Markets for Goods & Services

- ◆ Firms sell
- ◆ Households buy

Markets for Factors of Production

- ◆ Households sell
- ◆ Firms buy

The Circular-Flow Diagram

Factors of Production

- ◆ Inputs used to produce goods and services
- ◆ Land, labor, and capital

Cont'd

- Resources can be classified as **capital goods** and **labour**, although a more detailed classification would subdivide both these categories as
- (1) **Capital Goods:**
 - *Land*: all natural resources including agricultural land, rivers, forests, climate and mineral deposits;
 - *Capital*: all factories, offices, warehouses, shops, houses, machinery and equipment, stocks of materials, roads, railways, harbours, airports, vehicles, ships, schools, colleges, universities and similar assets produced in previous time periods;

Cont'd

- (2) **Labour Force:**
 - *Labour*, which is that proportion of the population able and willing to work, the quality being dependent upon variables such as age, effort, hours of work, education, skills, and attitudes;
 - *Enterprise*, which consists of those individuals in the labour force with organisational and managerial ability or with financial skills, such individuals taking risks in starting new companies and launching innovative products – sometimes with their own and sometimes with borrowed resources.

The Production Possibilities Frontier

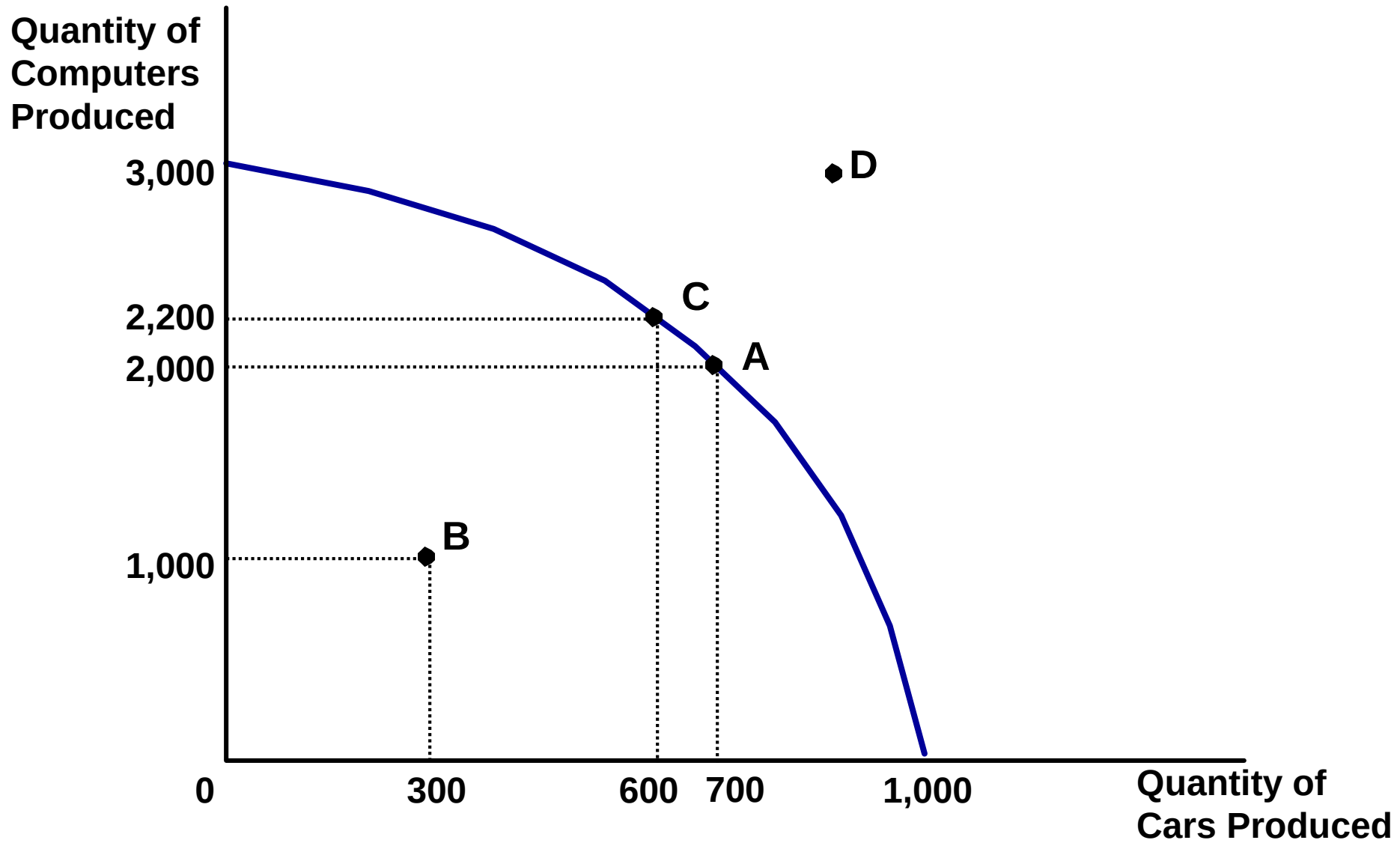
The production possibilities frontier is a graph showing the various combinations of output that the economy can possibly produce given the available factors of production and technology.

The Production Possibilities Frontier

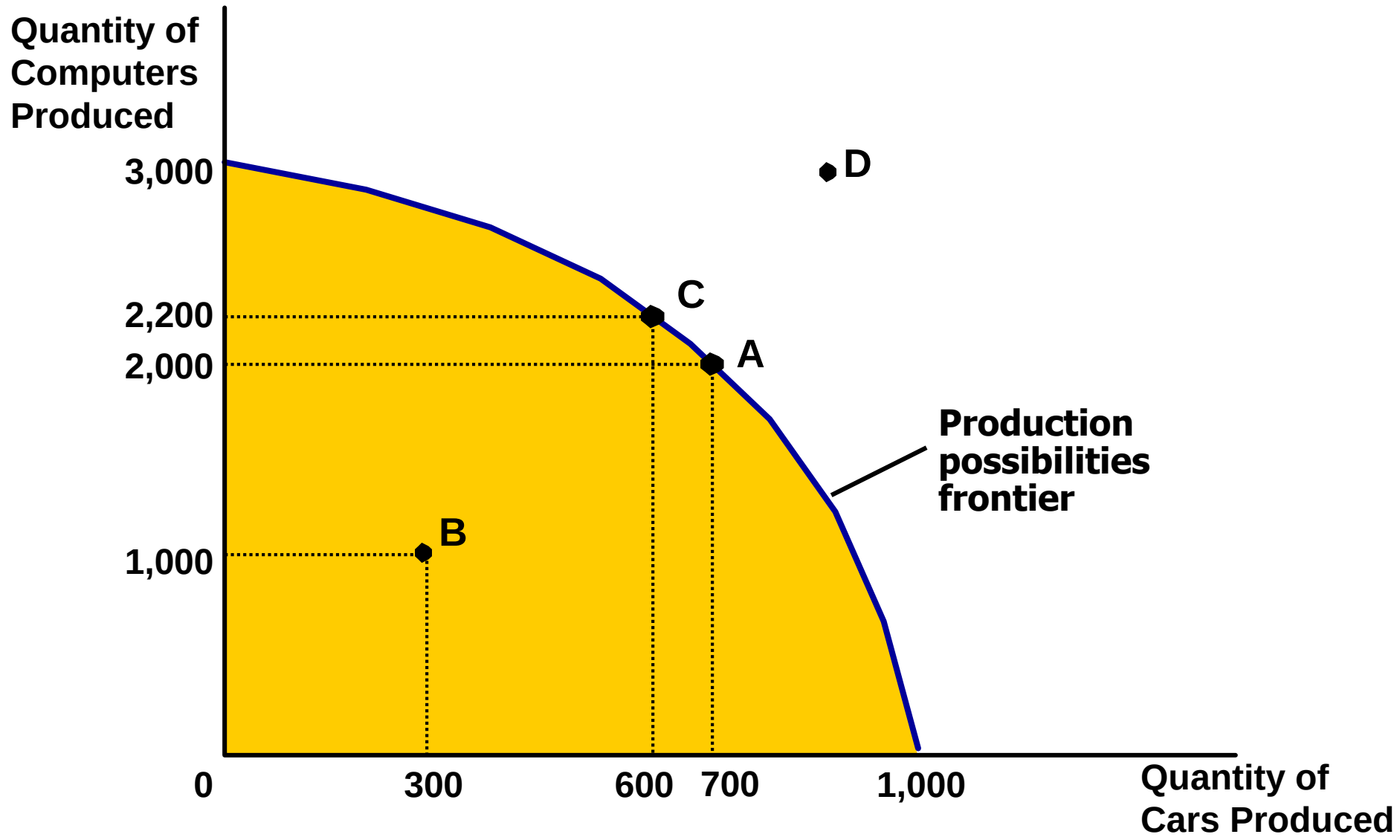
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- The concepts of scarcity, tradeoffs, choice, full employment, and efficiency are in most cases illustrated by Economists through the use of the notion of production possibilities curve (frontier).
- A **production possibility frontier (PPF)** is a curve or a boundary which shows the combinations of two or more goods and services that can be produced whilst using all of the available factor resources efficiently.

The Production Possibilities Frontier



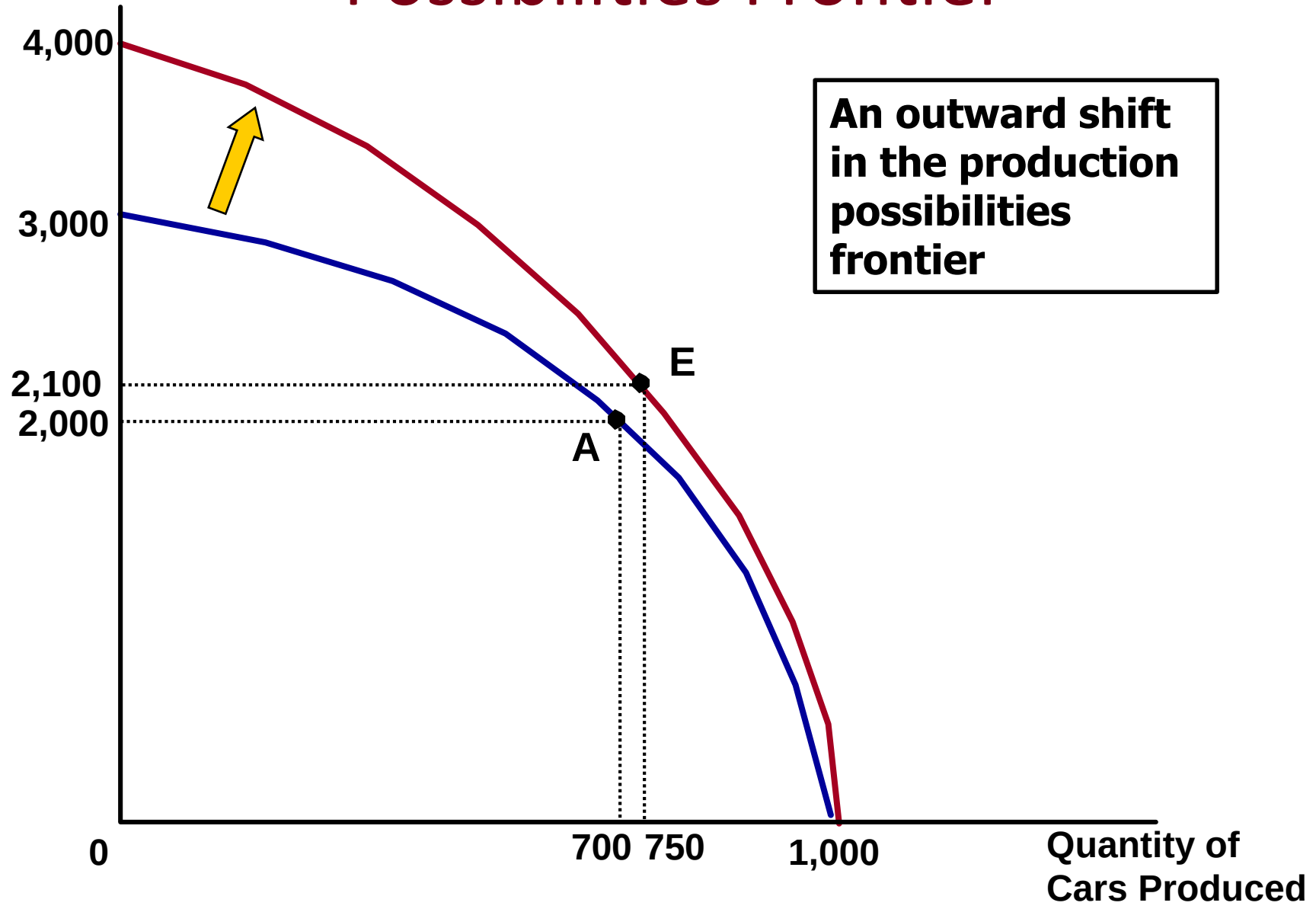
The Production Possibilities Frontier



Concepts Illustrated by the Production Possibilities Frontier

- ◆ Efficiency
- ◆ Tradeoffs
- ◆ Opportunity Cost
- ◆ Economic Growth

The Production Possibilities Frontier



Two Roles of Economists

- ◆ When they are trying to explain the world, they are *scientists*.
- ◆ When they are trying to change the world, they are *policymakers*.

Positive versus Normative Analysis

- ◆ Positive statements are statements that describe the world as it is.
 - ◆ Called *descriptive* analysis
- ◆ Normative statements are statements about how the world should be.
 - ◆ Called *prescriptive* analysis

Positive v. Normative Analysis

- Positive analysis
 - Addresses the economic consequences of a particular event or policy
 - Doesn't judge whether those consequences are desirable
- Normative analysis
 - Addresses the question of whether a policy should be used
 - Does involve the values of the person doing the analysis

Why Economists Disagree

- ◆ They may disagree on theories about how the world works.
- ◆ They may hold different values and, thus, different normative views.

Some Definitions

- **Gross national product (GNP)** is the value of the final goods and services produced. (total output)
- **Gross national expenditure (GNE)** is the value of total spending by the households. (total expenditure)
- **Gross national income (GNI)** is the value of the services of the factors of production hired by the firms (total income)

Opportunity Cost

- Opportunity Cost: The value of the next-best alternative that must be forgone in order to undertake an activity
 - Decisions depend upon opportunity costs
 - It is not the combined value of all other forgone activities, just the next best one

Opportunity cost Cont'd

- Opportunity cost is the **next best** alternative foregone...
- Potential output is the output that the economy would produce if both the labour force and the capital stock were fully employed.

Actual Output

- The actual output of an economy is known as the *gross national product (GNP)* ;
 - it is the value of all final goods and services produced in the economy in a year.
 - It is calculated by multiplying each good and service produced by its price and adding them all together, so that

$$GNP = \sum (ap_a + bp_b + \dots + zp_z)$$

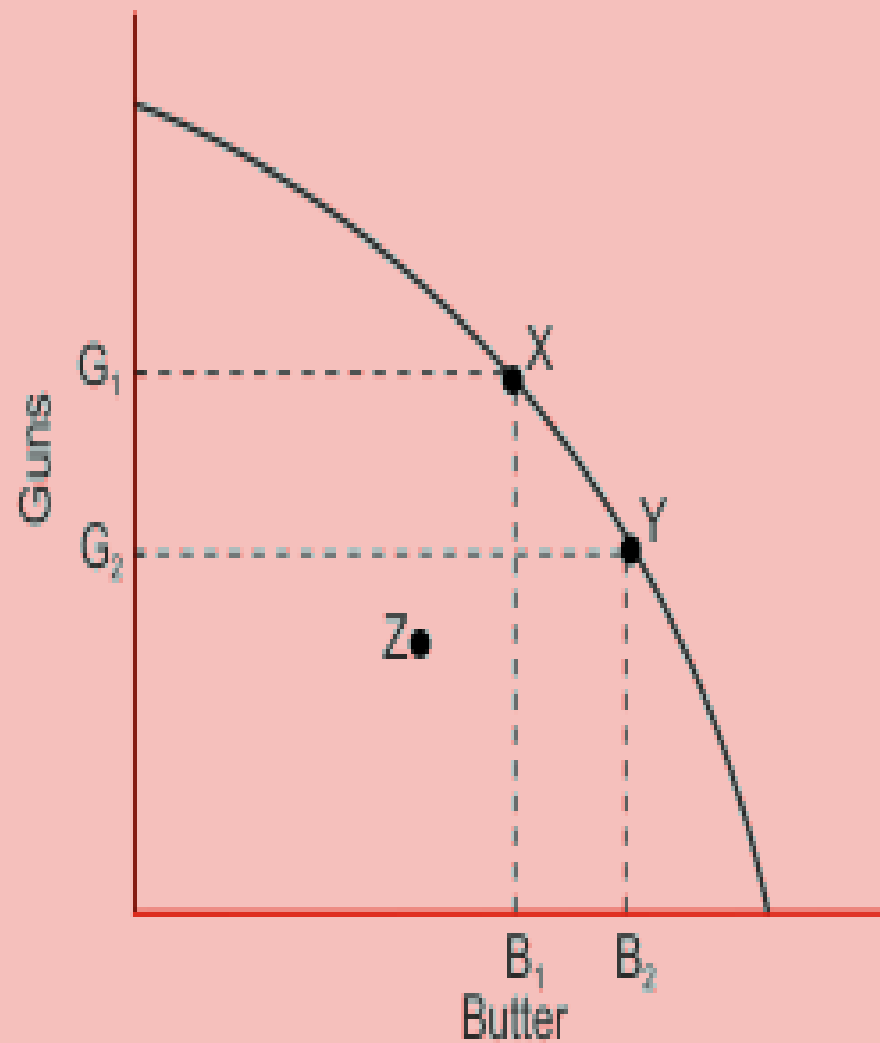
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- In the short run, however, say a year, a not unreasonable assumption in tackling macroeconomics is that the supply of factors of production and the state of technical knowledge are given or fixed.
- Thus it follows there must be a fixed upper limit to the total output that can be produced. This upper limit is the *productive potential* of the economy. This still leaves open the question as to whether potential output (Q) will in fact be equal to actual Gross National Product.
- It also still leaves open the question of how the composition of output is to be determined.

Cont'd

- What is important to note is that choosing one combination of outputs produced with a fixed quantity of inputs means forgoing alternative combinations. This is the well-known dilemma that all societies encounter: how to allocate scarce resources among competing uses.
- Consider a two-good economy where only guns and butter are produced, guns symbolising defence purposes and butter peaceful purposes. The production possibilities frontier sets the limit to the various combinations of the two goods that can be produced, if all resources are fully and efficiently employed (see [Figure 1](#)).

Figure 1 The production possibilities frontier in a two-good economy



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- The production possibilities frontier can be used to illustrate the microeconomic question of *what* should be produced. At a point *on* the frontier, such as X, the production of more butter, such as would be implied at Y, involves a sacrifice of some gun production.
- In other words, the additional output of butter at Y, given by B_1B_2 , has an opportunity cost.
- That opportunity cost is the gun output sacrificed by moving from X to Y, namely G_1G_2 .

Cont'd

- The example illustrates that microeconomics is concerned with explaining how choices are made, given that resources are limited relative to human wants.
- Clearly, if a society is on its production possibilities frontier, then the additional production of one good or service requires a reduction in the production of another good or service; in other words, when all resources are fully employed, additional output of one commodity requires the sacrifice of some other output.
- The opportunity cost involved is determined by the *slope* of the production possibilities frontier. Microeconomics is concerned with explaining how a society determines the allocation of resources between competing uses, given the fact of scarcity.

Costs Matter

- Economists count costs
 - Someone must pay them
- Scarcity is a fact of life
 - Never enough time, money, energy....
- Economics is the study of how people make choices under conditions of scarcity and of the results of those choices for society

Scarcity Principle

- Because of scarcity
 - Tradeoffs are widespread
 - Having more of one good usually means having less of another
 - AKA the “No free lunch Principle”

Cost-Benefit Principle

- Take an action if, and only if, the extra benefits from taking the action are at least as great as the extra costs
- Measuring the costs and benefits is often difficult
 - One may have to use assumptions

People Are Rational

- Economists assume that people are rational--that they try to fulfill their goals as best they can

Reservation Prices

- The highest price one would be willing to pay for any good or service
 - It is equal to the benefit received from the good or service

Economic Surplus

- The benefit of taking an action minus its cost
 - $\text{Economic Surplus} = \text{Benefit} - \text{Cost}$
 - Rational decision makers take all actions that yield a positive economic surplus

Role of Models

- An “abstract model” is a simplified description capturing essential elements of a situation
 - It allows logical analysis
 - It includes only the major forces at work and will ignore many details
- I.E., the cost-benefit principle is an abstract model of how an idealized individual would choose among competing alternatives

Recessions and Expansions

- Growth is not always steady
 - Recessions: Slowdowns in economic growth
 - Depressions: Severe recessions
 - Expansions: Periods of rapid economic growth

Unemployment

- Unemployment rate
 - The fraction of people who would like to be employed but can't find work
 - Key indicator of the state of the labor market

Inflation

- Inflation
 - The rate at which prices in general are increasing over time
 - Imposes a variety of costs on the economy
 - In recent years it has been relatively low

Measuring the Cost of Living

- **Inflation** refers to a situation in which the economy's overall price level is rising.
- The **inflation rate** is the percentage change in the price level from the previous period.

The Consumer Price Index

- The **consumer price index (CPI)** is a measure of the overall cost of the goods and services bought by a typical consumer.
- The **Central Statistical Office (CSO) [Zambian case]** reports the CPI each month.
 - Different countries have different names for the CSO
- It is used to monitor changes in the cost of living over time.

The Consumer Price Index

When the CPI rises, the typical family has to spend more kwachas (Dollars) to maintain the same standard of living.

How the Consumer Price Index Is Calculated

- **Fix the Basket:** Determine what prices are most important to the typical consumer.
 - ◆ The Central Statistical Office (CSO) identifies a market basket of goods and services the typical consumer buys.
 - ◆ The CSO conducts monthly consumer surveys to set the weights for the prices of those goods and services.

How the Consumer Price Index Is Calculated

- **Find the Prices:** Find the prices of each of the goods and services in the basket for each point in time.

How the Consumer Price Index Is Calculated

- **Compute the Basket's Cost:** Use the data on prices to calculate the cost of the basket of goods and services at different times.

How the Consumer Price Index Is Calculated

- Choose a Base Year and Compute the Index:
 - ◆ Designate one year as the base year, making it the benchmark against which other years are compared.
 - ◆ Compute the index by dividing the price of the basket in one year by the price in the base year and multiplying by 100.

How the Consumer Price Index Is Calculated

- Compute the inflation rate: The inflation rate is the percentage change in the price index from the preceding period.

The Inflation Rate

The **inflation rate** is calculated as follows:

$$\text{Inflation Rate in Year 2} = \frac{\text{CPI in Year 2} - \text{CPI in Year 1}}{\text{CPI in Year 1}} \times 100$$

Calculating the Consumer Price Index and the Inflation Rate: An Example

Step 1: Survey Consumers to Determine a Fixed Basket of Goods

4 hot dogs, 2 hamburgers

Calculating the Consumer Price Index and the Inflation Rate: An Example

Step 2: Find the Price of Each Good in Each Year

Year	Price of Hot dogs	Price of Hamburgers
2001	\$1	\$2
2002	\$2	\$3
2003	\$3	\$4

Calculating the Consumer Price Index and the Inflation Rate: An Example

Step 3: Compute the Cost of the Basket of Goods in Each Year

2001	$(\$1 \text{ per hot dog} \times 4 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 2 \text{ hamburgers}) = \textbf{\$8}$
2002	$(\$2 \text{ per hot dog} \times 4 \text{ hot dogs}) + (\$3 \text{ per hamburger} \times 2 \text{ hamburgers}) = \textbf{\$14}$
2003	$(\$3 \text{ per hot dog} \times 4 \text{ hot dogs}) + (\$4 \text{ per hamburger} \times 2 \text{ hamburgers}) = \textbf{\$20}$

Calculating the Consumer Price Index and the Inflation Rate: An Example

Step 4: Choose One Year as the Base Year (2001) and Compute the Consumer Price Index in Each Year

2001	$(\$8/\$8) \times 100 = \mathbf{100}$
2002	$(\$14/\$8) \times 100 = \mathbf{175}$
2003	$(\$20/\$8) \times 100 = \mathbf{250}$

Calculating the Consumer Price Index and the Inflation Rate: An Example

Step 5: Use the Consumer Price Index to Compute the Inflation Rate from Previous Year

2002	$(175-100)/100 \times 100 = \mathbf{75\%}$
2003	$(250-175)/175 \times 100 = \mathbf{43\%}$

Calculating the Consumer Price Index and the Inflation Rate: Another Example

- Base Year is 1998.
- Basket of goods in 1998 costs \$1,200.
- The same basket in 2000 costs \$1,236.
- $CPI = (\$1,236 / \$1,200) \times 100 = 103$.
- Prices increased 3 percent between 1998 and 2000.

GDP Deflator

The GDP deflator is calculated as follows:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Other Price Indexes

- Other prices indexes:
 - ◆ The producer price index, which measures the cost of a basket of goods and services bought by firms rather than consumers.

Problems in Measuring The Cost of Living

The CPI is an accurate measure of the selected goods that make up the typical bundle, but it is not a perfect measure of the cost of living.

Problems in Measuring The Cost of Living

- Substitution bias
- Introduction of new goods
- Unmeasured quality changes

Substitution Bias

- The basket does not change to reflect consumer reaction to changes in relative prices.
 - ◆ Consumers substitute toward goods that have become relatively less expensive.
 - ◆ The index overstates the increase in cost of living by not considering consumer substitution.

Introduction of New Goods

- The basket does not reflect the change in purchasing power brought on by the introduction of new products.
 - ◆ New products result in greater variety, which in turn makes each dollar more valuable.
 - ◆ Consumers need fewer dollars to maintain any given standard of living.

Unmeasured Quality Changes

- If the quality of a good rises from one year to the next, the value of a kwacha rises, even if the price of the good stays the same.
- If the quality of a good falls from one year to the next, the value of a kwacha falls, even if the price of the good stays the same.

Unmeasured Quality Changes

The CSO tries to adjust the price for constant quality, but such differences are hard to measure.

Problems in Measuring the Cost of Living

- The substitution bias, introduction of new goods, and unmeasured quality changes cause the CPI to overstate the true cost of living.

The GDP Deflator versus the Consumer Price Index

- Economists and policymakers monitor both the GDP deflator and the consumer price index to gauge how quickly prices are rising.
- There are two important differences between the indexes that can cause them to diverge.

The GDP Deflator versus the Consumer Price Index

- The **GDP deflator** reflects the prices of all goods and services produced domestically, whereas...
- ...the **consumer price index** reflects the prices of all goods and services bought by consumers.

The GDP Deflator versus the Consumer Price Index

- The **consumer price index** compares the price of a *fixed basket* of goods and services to the price of the basket in the base year (only occasionally does the CSO change the basket)...
- ...whereas the **GDP deflator** compares the price of *currently produced* goods and services to the price of the same goods and services in the base year.

Dollar Figures from Different Times

Price indexes are used to correct for the effects of inflation when comparing Dollar figures from different times.

Dollar Figures from Different Times

- Do the following to convert (inflate) wages in 1931 to dollars in 1999:

$$\text{Salary}_{1999} = \text{Salary}_{1931} \times \frac{\text{Price level in 1999}}{\text{Price level in 1931}}$$

Dollar Figures from Different Times

- Do the following to convert (inflate) wages in 1931 to Kwachas in 1999:

$$\text{Salary}_{1999} = \text{Salary}_{1931} \times \frac{\text{Price level in 1999}}{\text{Price level in 1931}}$$

$$= \$80,000 \times \frac{166}{15.2}$$

$$= \$873,684$$

Indexation

When some Kwacha amount is automatically corrected for inflation by law or contract the amount is said to be indexed for inflation.

Real and Nominal Interest Rates

Interest represents a payment in the future for a transfer of money in the past.

Real and Nominal Interest Rates

- The **nominal interest rate** is the interest rate not corrected for inflation.
 - ◆ It is the interest rate that a bank pays.
- The **real interest rate** is the nominal interest rate that is corrected for inflation.

$$\text{Real interest rate} = (\text{Nominal interest rate} - \text{Inflation rate})$$

Real and Nominal Interest Rates

- You borrowed \$1,000 for one year.
- Nominal interest rate was 15%.
- During the year inflation was 10%.

$$\begin{aligned}\text{Real interest rate} &= \text{Nominal interest rate} - \text{Inflation} \\ &= 15\% - 10\% = 5\%\end{aligned}$$

Different economic systems

Three types of economic systems to answer the what, how and for whom?

- 1) *Planned/command economies* – a small group of individuals (Central planning committee) make the major economic decisions, (features – 5 year plans for various sectors of the economy; Wage levels and prices set by planners and when fail to work Surpluses or shortages (long queues result and black markets form) emerge.
- 2) *Market(capitalist/free enterprise) economies* – the economic decision is made through the market system through the interaction of buyers and sellers. (i.e. the forces of demand and Supply)
- 3) **Mixed Economy**- Combination of the above two(2)

The Economy's Income and Expenditure

When judging whether the economy is doing well or poorly, it is natural to look at the total income that everyone in the economy is earning.

The Economy's Income and Expenditure

- For an economy as a whole, income must equal expenditure because:
 - ◆ Every transaction has a buyer and a seller.
 - ◆ Every dollar of spending by some buyer is a dollar of income for some seller.

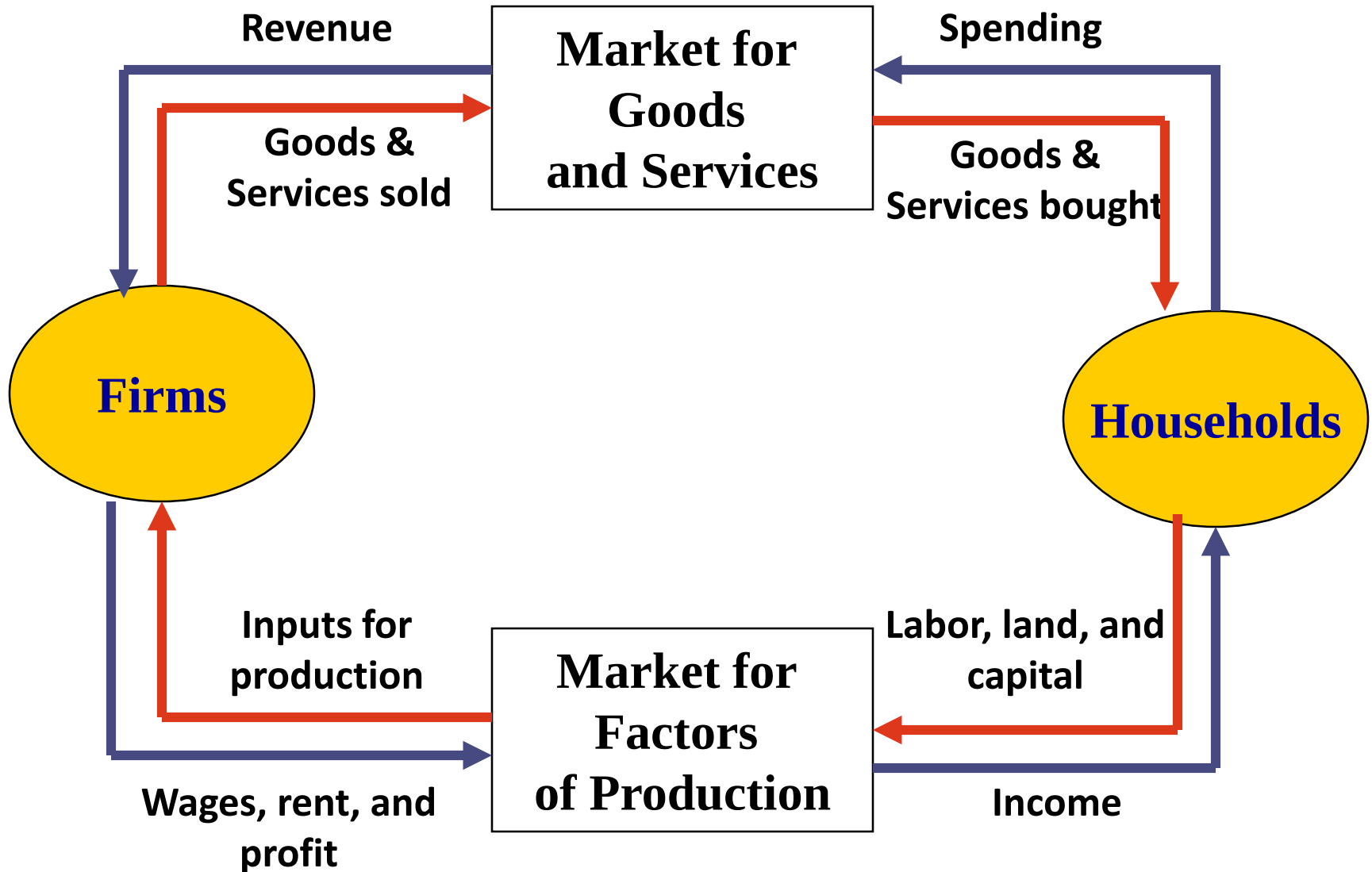
Gross Domestic Product

- Gross domestic product (GDP) is a measure of the income and expenditures of an economy.
- It is the total market value of all final goods and services produced within a country in a given period of time.

The Circular-Flow Diagram

The equality of income and expenditure can be illustrated with the **circular-flow diagram**.

The Circular-Flow Diagram



The Measurement of GDP

GDP is the market value of all final goods and services produced within a country in a given period of time.

The Measurement of GDP

- Output is valued at market prices.
- It records only the value of final goods, not intermediate goods (the value is counted only once).
- It includes both tangible goods (food, clothing, cars) and intangible services (haircuts, housecleaning, doctor visits).

The Measurement of GDP

- It includes goods and services **currently produced**, not transactions involving goods produced in the past.
- It measures the value of production **within the geographic confines of a country**.

The Measurement of GDP

- It measures the value of production that takes place within a specific interval of time, usually a year or a quarter (three months).

What Is Counted in GDP?

GDP includes all items
produced in the economy
and sold legally in markets.

What Is Not Counted in GDP?

- GDP excludes most items that are produced and consumed at home and that never enter the marketplace.
- It excludes items produced and sold illicitly, such as illegal drugs.

Other Measures of Income

- 1) Gross National Product (GNP)
- 2) Net National Product (NNP)
- 3) National Income
- 4) Personal Income
- 5) Disposable Personal Income

1) Gross National Product

- Gross national product (GNP) is the total income earned by a nation's permanent residents (called nationals).
- It differs from GDP by including income that our citizens earn abroad and excluding income that foreigners earn here.

2) Net National Product (NNP)

- Net National Product (NNP) is the total income of the nation's residents (GNP) minus losses from depreciation.
- Depreciation is the wear and tear on the economy's stock of equipment and structures.

3) National Income

- **National Income** is the total income earned by a nation's residents in the production of goods and services.
- It differs from NNP by excluding **indirect business taxes** (such as sales taxes) and including business subsidies.

4) Personal Income

- **Personal income** is the income that households and noncorporate businesses receive.
- Unlike national income, it excludes **retained earnings**, which is income that corporations have earned but have not paid out to their owners.
- In addition, it includes household's **interest income** and government **transfers**.

5) Disposable Personal Income

- **Disposable personal income** is the income that household and noncorporate businesses have left after satisfying all their obligations to the government.
- It equals personal income minus personal taxes and certain nontax payments.

The Components of GDP

GDP (Y) is the sum of the following:

- ◆ Consumption (C)
- ◆ Investment (I)
- ◆ Government Purchases (G)
- ◆ Net Exports (NX)

$$Y = C + I + G + NX$$

The Components of GDP

- Consumption (C):
 - ◆ The spending by households on goods and services, with the exception of purchases of new housing.
- Investment (I):
 - ◆ The spending on capital equipment, inventories, and structures, including new housing.

The Components of GDP

- **Government Purchases (G):**
 - ◆ The spending on goods and services by local, state, and federal governments.
 - ◆ Does not include transfer payments because they are not made in exchange for currently produced goods or services.
- **Net Exports (NX):**
 - ◆ Exports minus imports.

Real versus Nominal GDP

- **Nominal GDP** values the production of goods and services at current prices.
- **Real GDP** values the production of goods and services at constant prices.

Real versus Nominal GDP

An accurate view of the economy requires adjusting nominal to real GDP by using the GDP deflator.

GDP Deflator

- The **GDP deflator** measures the current level of prices relative to the level of prices in the base year.
- It tells us the rise in nominal GDP that is attributable to a rise in prices rather than a rise in the quantities produced.

GDP Deflator

The GDP deflator is calculated as follows:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Converting Nominal GDP to Real GDP

Nominal GDP is converted to real GDP as follows:

$$\text{Real GDP}_{20xx} = \frac{(\text{Nominal GDP}_{20xx})}{(\text{GDP deflator}_{20xx})} \times 100$$

Real and Nominal GDP

Year	Price of Hot dogs	Quantity of Hot dogs	Price of Hamburgers	Quantity of Hamburgers
2001	\$1	100	\$2	50
2002	\$2	150	\$3	100
2003	\$3	200	\$4	150

Real and Nominal GDP

Calculating Nominal GDP:

2001	$(\$1 \text{ per hot dog} \times 100 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 50 \text{ hamburgers}) = \textbf{\$200}$
2002	$(\$2 \text{ per hot dog} \times 150 \text{ hot dogs}) + (\$3 \text{ per hamburger} \times 100 \text{ hamburgers}) = \textbf{\$600}$
2003	$(\$3 \text{ per hot dog} \times 200 \text{ hot dogs}) + (\$4 \text{ per hamburger} \times 150 \text{ hamburgers}) = \textbf{\$1200}$

Real and Nominal GDP

Calculating Real GDP (base year 2001):

2001	$(\$1 \text{ per hot dog} \times 100 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 50 \text{ hamburgers}) = \text{\textcolor{blue}{\$200}}$
2002	$(\$1 \text{ per hot dog} \times 150 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 100 \text{ hamburgers}) = \text{\textcolor{blue}{\$350}}$
2003	$(\$1 \text{ per hot dog} \times 200 \text{ hot dogs}) + (\$2 \text{ per hamburger} \times 150 \text{ hamburgers}) = \text{\textcolor{blue}{\$500}}$

Real and Nominal GDP

Calculating the GDP Deflator:

2001	$(\$200/\$200) \times 100 = \mathbf{100}$
2002	$(\$600/\$350) \times 100 = \mathbf{171}$
2003	$(\$1200/\$500) \times 100 = \mathbf{240}$

GDP and Economic Well-Being

- GDP is the best single measure of the economic well-being of a society.
- GDP per person tells us the income and expenditure of the average person in the economy.

GDP and Economic Well-Being

- Higher GDP per person indicates a higher standard of living.
- GDP is not a perfect measure of the happiness or quality of life, however.

GDP and Economic Well-Being

- Some things that contribute to well-being are not included in GDP.
 - ◆ The value of leisure.
 - ◆ The value of a clean environment.
 - ◆ The value of almost all activity that takes place outside of markets, such as the value of the time parents spend with their children and the value of volunteer work.